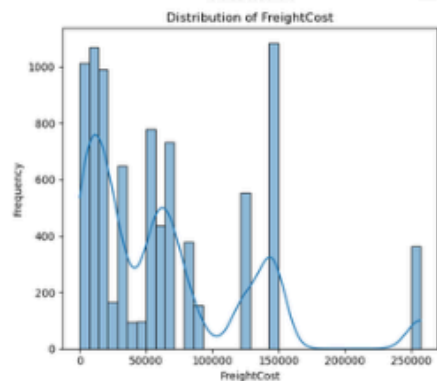
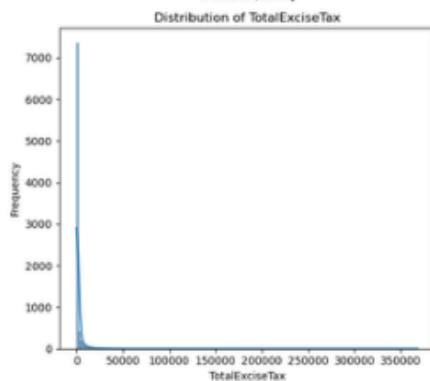
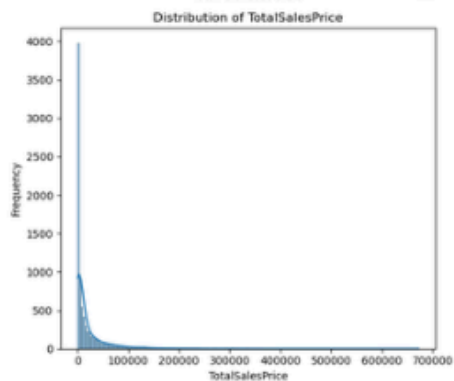
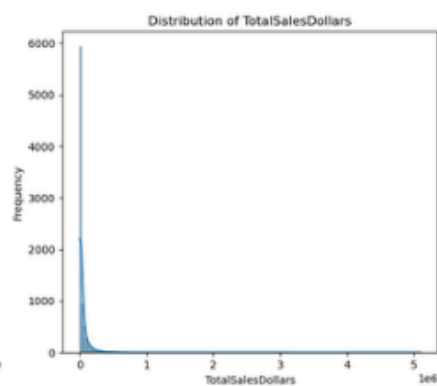
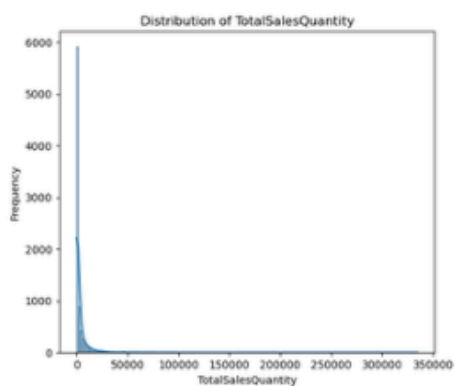
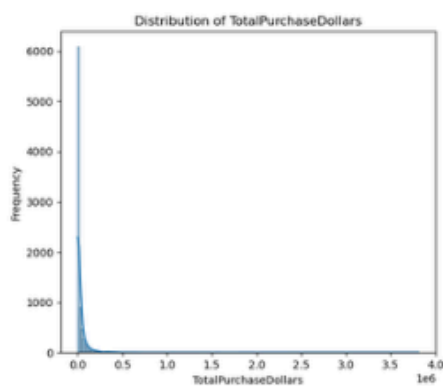
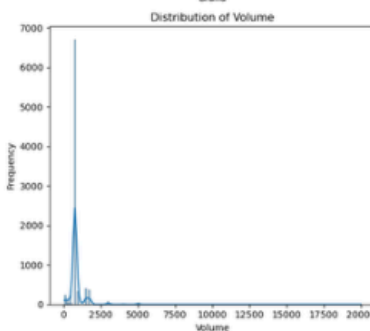
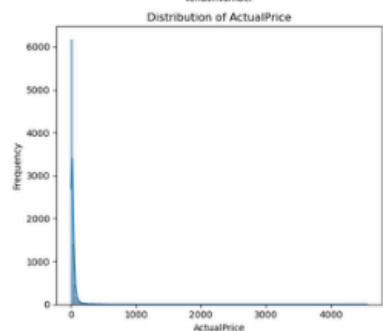
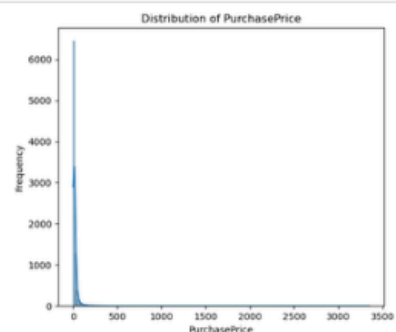
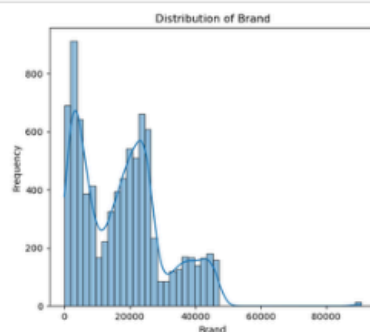
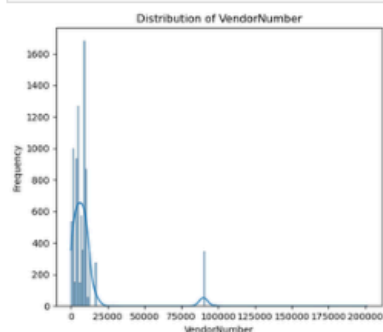


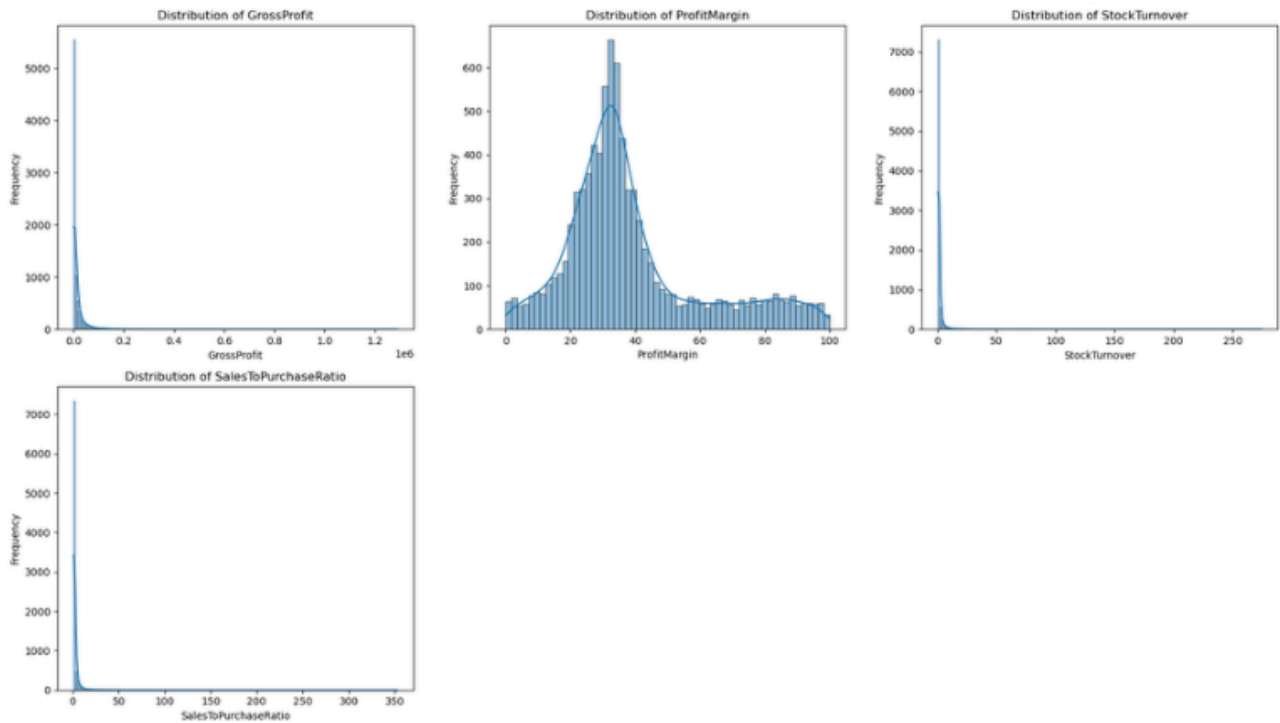
Inventory & Vendor Performance Analytic

Exploratory Data Analysis

Summary Statistics

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10693.00	10649.89	18752.81	2.00	3951.00	7153.00	9552.00	201359.00
Brand	10693.00	18037.74	12662.53	58.00	5789.00	18761.00	25514.00	90631.00
PurchasePrice	10693.00	24.38	109.26	0.00	6.84	10.45	19.47	5681.81
ActualPrice	10693.00	35.64	148.24	0.00	10.99	15.99	28.99	7499.99
Volume	10693.00	847.35	664.28	50.00	750.00	750.00	750.00	20000.00
TotalPurchaseQuantity	10693.00	3140.78	11094.57	1.00	36.00	262.00	1978.00	337660.00
TotalPurchaseDollars	10693.00	30103.88	123062.39	0.00	453.18	3655.23	20733.42	3811251.60
TotalSalesQuantity	10693.00	3077.36	10952.35	0.00	33.00	261.00	1929.00	334939.00
TotalSalesDollars	10693.00	42244.31	167648.30	0.00	729.27	5298.21	28406.05	5101919.51
TotalSalesPrice	10693.00	18797.69	44952.48	0.00	289.71	2857.80	16075.38	672819.31
TotalExciseTax	10693.00	1774.19	10975.07	0.00	4.80	46.59	419.16	368242.80
FreightCost	10693.00	61428.92	60937.67	0.09	14069.87	50293.62	79528.99	257032.07
GrossProfit	10693.00	12140.43	46229.68	-52002.78	52.92	1399.73	8661.82	1290667.91
ProfitMargin	10693.00	-inf	NaN	-inf	13.33	30.41	39.96	100.00
StockTurnover	10693.00	1.71	6.02	0.00	0.81	0.98	1.04	274.50
SalesToPurchaseRatio	10693.00	inf	NaN	0.00	1.15	1.44	1.67	inf





Summary Statistics Insights

The summary statistics provide an initial understanding of the distribution, variability, and overall characteristics of the key business metrics. They also highlight potential anomalies that warrant further investigation during the exploratory data analysis.

Negative and Zero Values

- **Gross Profit:** The minimum gross profit recorded is **-\$52,002.78**, indicating that certain products or transactions generated losses. This may be attributed to high procurement costs, aggressive discounting, or pricing strategies that resulted in selling below cost.
- **Profit Margin:** The minimum value is **negative infinity ($-\infty$)**, which occurs when sales revenue is zero while procurement costs are incurred. Such observations represent exceptional

cases and should be interpreted carefully in profitability analyses.

- **Sales Quantity and Sales Revenue:** Both metrics contain observations with values of **zero**, indicating that some products were purchased but not sold during the analysis period. These items may represent **slow-moving, newly introduced**, or **obsolete inventory**, leading to capital being tied up in stock.
-

High Variability and Potential Outliers

Several numerical variables exhibit substantial variation, suggesting the presence of outliers and premium-value products.

- **Purchase Price and Actual Selling Price:** The maximum purchase price (**\$5,681.81**) and selling price (**\$7,499.99**) are considerably higher than their respective mean values (**\$24.39** and **\$35.64**). This indicates the existence of premium or luxury products within the inventory portfolio.
- **Freight Cost:** Freight expenses range from **\$0.09** to **\$257,032.07**, reflecting significant differences in shipping costs. This variation may result from bulk procurement, shipment size, transportation distance, or vendor-specific logistics arrangements.
- **Stock Turnover:** Stock turnover values range from **0** to **274.50**, demonstrating considerable differences in inventory movement across products. A turnover ratio of **0** indicates products with no recorded sales, whereas values **greater than 1** suggest that sales exceeded purchases during the analysis period, likely because demand was fulfilled using inventory carried over from previous stock periods.

Data Filtering

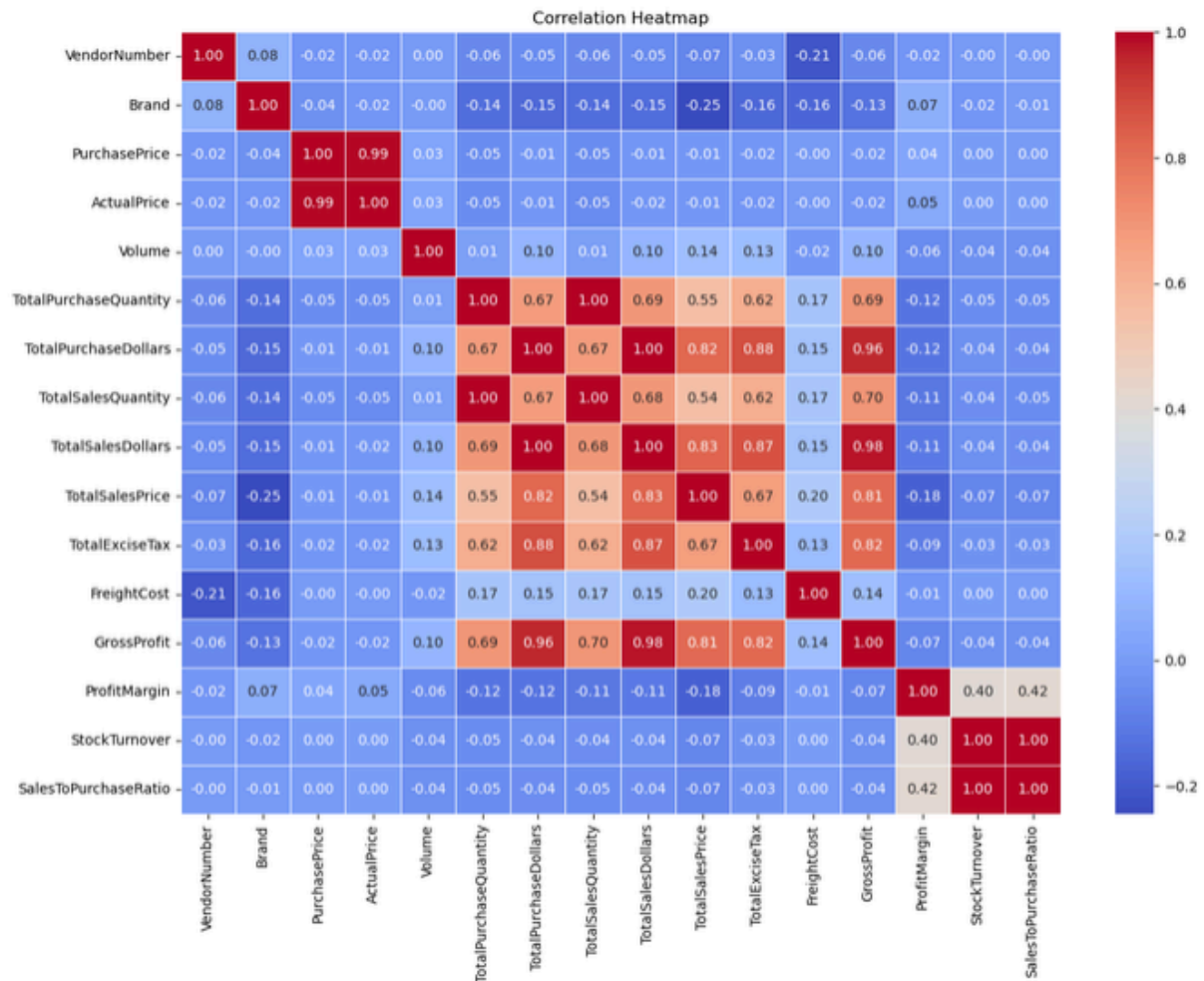
The following filtering operations were performed to prepare the data for analysis:

- **Excluded invalid purchase records** by retaining only records where **PurchasePrice > 0**, ensuring that pricing analyses were based on valid purchase transactions.
- **Filtered missing sales records** by identifying products purchased but not sold, which helped detect **slow-moving and unsold inventory**.
- **Filtered vendors based on sales performance** using the **75th percentile (top-performing vendors)** and **25th percentile (low-performing vendors)** of TotalSalesDollars for statistical analysis.
- **Filtered products with Stock Turnover < 1** to identify vendors with **slow-moving inventory** and potential excess stock.
- **Filtered the Top 10 vendors** based on TotalPurchaseDollars to analyze procurement dependency using **Pareto Analysis**.
- **Filtered brands with high profit margins but low sales performance** to identify products that could benefit from pricing adjustments or promotional campaigns.
- **Filtered vendor-brand combinations** during SQL joins to retain only matching purchase records while preserving unmatched sales data using appropriate SQL joins.
- **Filtered negative unsold inventory values** (where applicable) by restricting unsold inventory to non-negative values before calculating capital locked in inventory.
- **Filtered specific vendors and brands** during data validation to investigate missing PurchasePrice values and verify join consistency across tables.

Purpose of Data Filtering

These filtering operations ensured that the dataset was clean, analytically consistent, and focused on records relevant to procurement, inventory, sales, profitability, and vendor performance analysis. They also enabled accurate feature engineering, exploratory data analysis, and statistical testing.

Correlation Insights



Correlation analysis was performed to examine the relationships between key procurement, sales, inventory, and profitability metrics. The findings provide valuable insights into the factors influencing business performance.

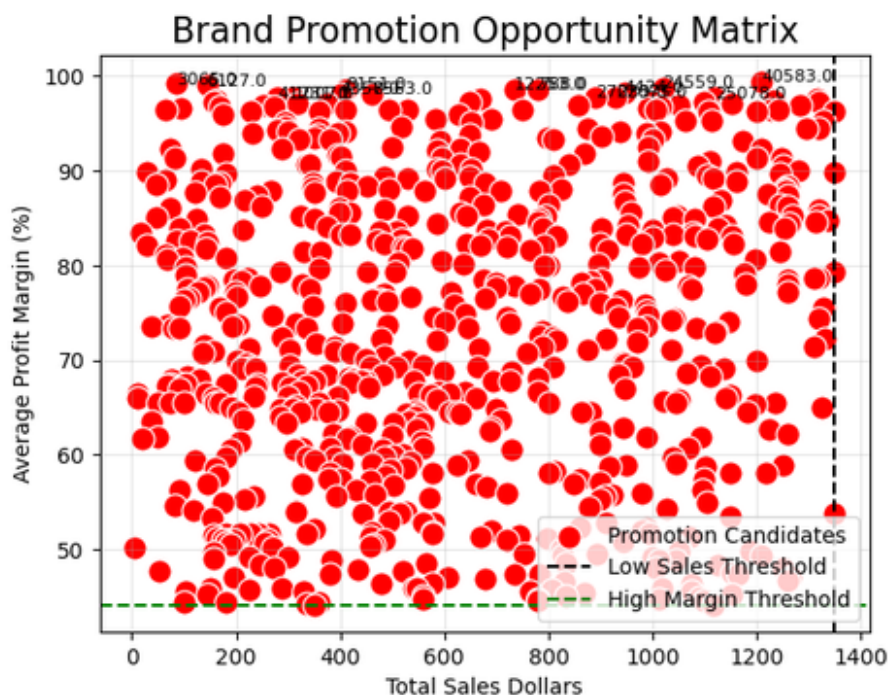
Key Insights

- Purchase Price vs. Sales Performance:** Purchase Price exhibits a **very weak negative correlation** with both **Total Sales Dollars (-0.012)** and **Gross Profit (-0.016)**. This indicates that variations in procurement cost have minimal influence on overall sales revenue and profitability, suggesting that other factors such as demand, product mix, and pricing strategies play a more significant role.

- **Purchase Quantity vs. Sales Quantity:** A near-perfect positive correlation (0.999) exists between **Total Purchase Quantity** and **Total Sales Quantity**. This demonstrates that inventory procurement closely aligns with customer demand, reflecting efficient inventory planning and stock replenishment practices.
- **Profit Margin vs. Total Sales Price:** Profit Margin shows a **weak negative correlation (-0.179)** with **Total Sales Price**. This suggests that products generating higher sales revenue do not necessarily achieve higher profit margins. Competitive pricing strategies, promotional discounts, or increased procurement costs may contribute to reduced margins despite higher sales.
- **Stock Turnover vs. Profitability:** Stock Turnover exhibits **weak negative correlations** with both **Gross Profit (-0.038)** and **Profit Margin (-0.055)**. These results indicate that faster inventory movement alone does not guarantee improved profitability. Inventory turnover should therefore be evaluated alongside pricing, procurement costs, and product margins.

Key Findings:-

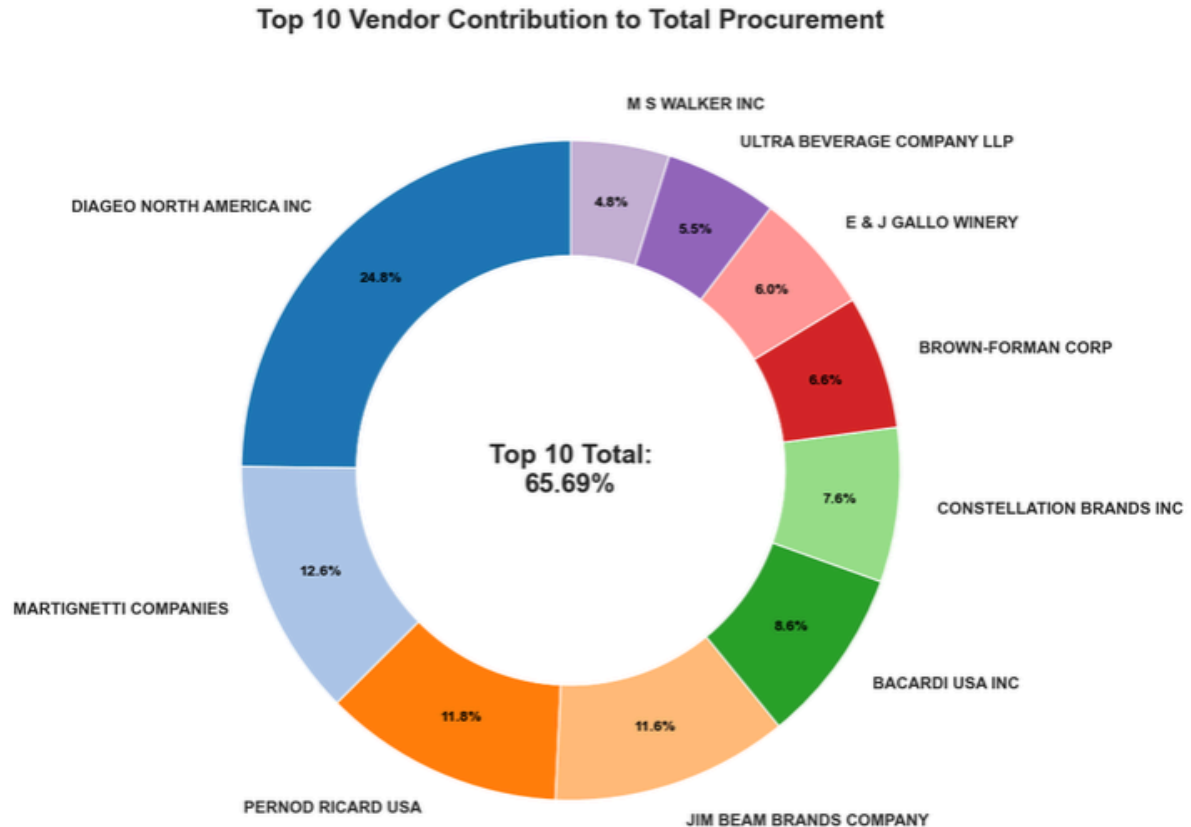
1- brands which needs Promotional and Pricing Adjustment



	Brand	TotalSalesQuantity	TotalSalesDollars	ProfitMargin
7957	40583	110	1208.90	99.39
1209	3065	86	85.14	99.17
2313	6127	72	143.28	98.97
6277	24559	51	1019.49	98.68
2725	8151	415	410.85	98.62
3301	12253	92	735.08	98.58
270	788	52	779.48	98.56
1866	4424	86	945.14	98.22
1835	4357	40	399.60	98.13
2878	8563	42	461.58	98.11
5480	22075	31	929.69	97.81
7012	27033	27	890.73	97.78
412	1007	31	309.69	97.68
8002	41231	20	279.80	97.67
6466	25078	28	1119.72	97.67
1241	3152	61	670.39	97.64
1933	4723	1330	1316.70	97.60
4172	17172	27	161.73	97.46
3888	15871	77	1231.23	97.33
6428	24994	25	649.75	97.32

A total of 198 brands were identified as having low sales performance despite maintaining high profit margins. These brands represent potential opportunities for targeted marketing campaigns, promotional strategies, or pricing optimization to increase sales volume while preserving profitability.

2- total procurement is dependent on the top vendors



The **top 10 vendors account for approximately 65.69% of the total procurement value**, indicating a high concentration of purchasing among a limited number of suppliers. **DIAGEO NORTH AMERICA INC** contributes the largest share (24.8%), followed by **MARTIGNETTI COMPANIES** (12.6%) and **PERNOD RICARD USA** (11.8%). This level of supplier dependency may expose the organization to supply chain risks, highlighting the importance of supplier diversification and strategic procurement planning.

3- Impact on Bulk purchasing on Cost saving

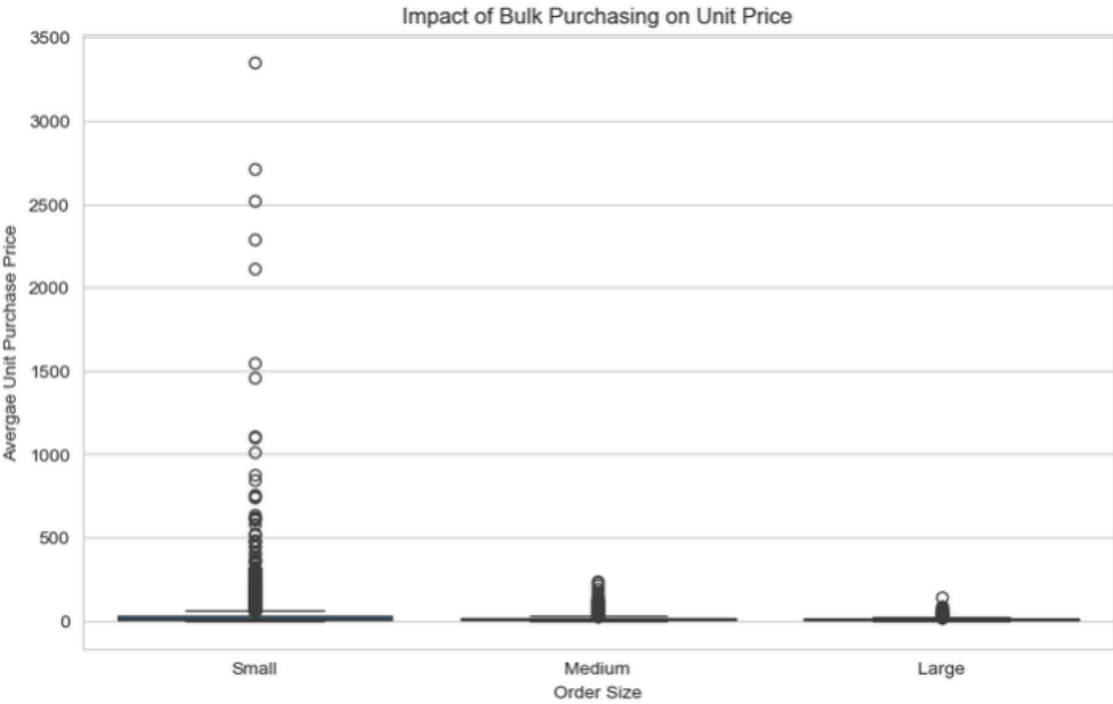
- Vendors purchasing in bulk (large order sizes) receive the lowest average unit purchase price (approximately \$10.78 per unit), indicating greater opportunities for cost savings and improved profit margins when inventory is managed efficiently.
- The difference in unit price between small and large orders is substantial (approximately 72% lower for large orders), demonstrating a significant volume-based pricing advantage.

- These findings suggest that the organization's bulk purchasing strategy effectively incentivizes larger order quantities, enabling vendors to reduce procurement costs while potentially increasing overall profitability, despite lower revenue on a per-unit basis.

]: UnitPurchasePrice

OrderSize

Small	39.07
Medium	15.48
Large	10.77



4- Vendors with low Inventory turnover

Total Unsold Inventory Capital: \$2.71M

Slow-moving inventory increases storage costs, reduces cash flow efficiency, and affects overall profitability.

Identifying vendors with low inventory turnover enables better stock management, minimizing financial strain.

	VendorName	UnsoldInventoryValue
25	DIAGEO NORTH AMERICA INC	\$0.72M
46	JIM BEAM BRANDS COMPANY	\$0.55M
68	PERNOD RICARD USA	\$0.47M
116	WILLIAM GRANT & SONS INC	\$0.40M
30	E & J GALLO WINERY	\$0.23M
79	SAZERAC CO INC	\$0.20M
11	BROWN-FORMAN CORP	\$0.18M
20	CONSTELLATION BRANDS INC	\$0.13M
61	MOET HENNESSY USA INC	\$0.13M
77	REMY COINTREAU USA INC	\$0.12M

11]: StockTurnover

VendorName	
ALISA CARR BEVERAGES	0.62
HIGHLAND WINE MERCHANTS LLC	0.71
PARK STREET IMPORTS LLC	0.75
Circa Wines	0.76
Dunn Wine Brokers	0.77
CENTEUR IMPORTS LLC	0.77
SMOKY QUARTZ DISTILLERY LLC	0.78
TAMWORTH DISTILLING	0.80
THE IMPORTED GRAPE LLC	0.81
WALPOLE MTN VIEW WINERY	0.82

5- Profit Margin Comparison : High vs Low - profit margin

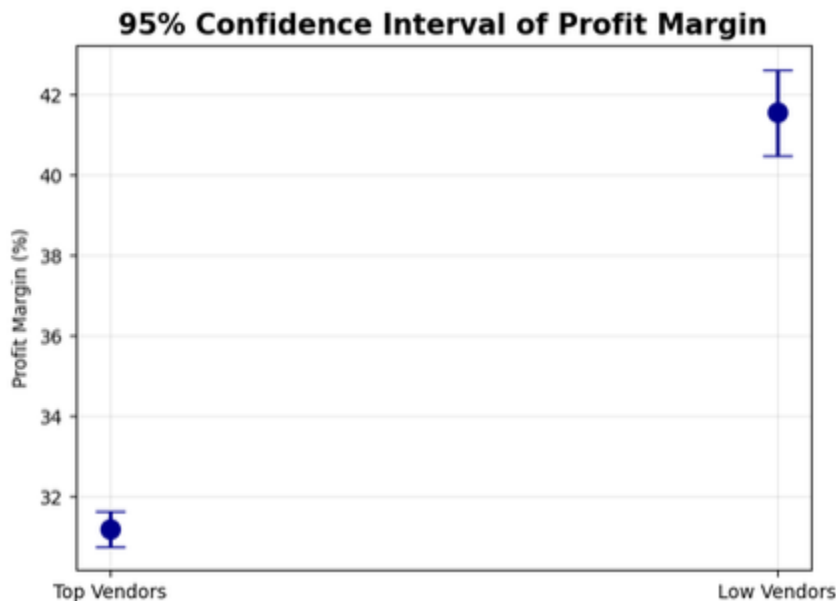
The **low-performing vendors** exhibit a **higher average profit margin (~41.5%)** compared to the **top-performing vendors (~31.2%)**.

The **95% confidence intervals do not overlap**, indicating that the difference in average profit margins between the two groups is statistically significant.

Although top-performing vendors generate substantially higher sales volumes, they operate with relatively **lower profit margins**, suggesting a high-volume, lower-margin business strategy.

In contrast, low-performing vendors generate fewer sales but maintain **higher profit margins**, indicating they may be selling premium or niche products with greater per-unit profitability.

This analysis demonstrates that **higher sales do not necessarily translate into higher profit margins**. Revenue and profitability should therefore be evaluated together when assessing vendor performance.



6- Statistical Validation of Profit Margin Difference

Hypothesis:

H₀ (Null Hypothesis): There is **no significant difference** in the mean profit margins of top-performing and low-performing vendors.

H₁ (Alternative Hypothesis): The mean profit margins of top-performing and low-performing vendors are **significantly different**.

T-statistic : -17.5815

P-value : 0.0000

Reject the Null Hypothesis (H₀)

There is a statistically significant difference in the mean profit margins between top-performing and low-performing vendors.

Final Recommendations

Based on the findings of this analysis, the following recommendations are proposed to improve procurement efficiency, inventory management, and overall business performance.

1. Diversify Procurement Across Vendors

The analysis revealed that the **top 10 vendors account for approximately 65.69% of total procurement expenditure**, indicating a high dependency on a limited number of suppliers. Expanding the supplier base can reduce supply chain risks, improve bargaining power, and enhance procurement resilience.

2. Optimize Inventory Management

Approximately **\$2.71 million** of capital is tied up in unsold inventory. The company should implement improved demand forecasting, inventory monitoring, and replenishment strategies to reduce excess stock and improve inventory turnover.

3. Promote High-Margin, Low-Sales Products

The analysis identified **198 brands** with relatively low sales but high profit margins. Targeted marketing campaigns, promotional offers, and optimized pricing strategies can help increase sales volume while maintaining profitability.

4. Leverage Bulk Purchasing for High-Demand Products

The pricing analysis indicates that larger purchase volumes are generally associated with lower unit purchase costs. The organization should continue utilizing bulk procurement for products with stable demand while avoiding overstocking of slow-moving items.

5. Monitor Vendor Performance Using Multiple KPIs

Vendor evaluation should not rely solely on sales revenue. Performance should be assessed using a combination of **Gross Profit, Profit Margin, Inventory Turnover, Sales-to-Purchase Ratio, Procurement Contribution, and Freight Cost** to support more informed procurement decisions.

6. Improve Freight Cost Efficiency

The significant variation observed in freight costs suggests opportunities to optimize logistics. Consolidating shipments, negotiating transportation contracts, and reviewing vendor-specific freight arrangements can help reduce procurement costs.

7. Reduce Supplier Dependency Risk

Given the concentration of procurement among a few vendors, the organization should develop contingency sourcing strategies and strengthen relationships with alternative suppliers to improve supply chain resilience.

8. Establish KPI Dashboards for Continuous Monitoring

Implementing interactive dashboards to monitor procurement spend, vendor performance, inventory turnover, profitability, and capital locked in inventory will enable management to make timely, data-driven decisions and respond proactively to operational challenges.

